

Pavel Petráček

PhD in Autonomous Flying Robotics
Robotics engineer & team lead

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About me

Robotics R&D engineer and product owner with 11 years of hands-on, real-world experience in fully-autonomous robotics: indoor UAVs and multi-robot systems. Experienced in owning robotics projects from concept to deployment in high-impact environments of subterranean S&R, indoor flying, and others. Technical expert with strong academic background, team leadership and project management.

Fast learner open to new things.

Core skills

GNSS-denied autonomy
Perception, SLAM, 3D mapping
Sensor fusion (LiDAR, visual, IMU)
Multi-agent systems
Technical and R&D leadership
Systems engineering

- C++, Python, ROS, PX4, git
- Docker, Singularity, Gazebo
- RTOS/embedded targets

Languages

Czech (native)
English (fluent)

Experience

Independent development for startups *1/2026–present*

- RoboTwin (CZ): perception for no-code teaching and programming of industrial robots

Fly4Future s.r.o. (startup) *4/2023–12/2025*
R&D Projects Lead & Hands-On Developer

- Product owner of **RoboAid**: autonomous UAV inspections
- Team lead of 6 roboticists; management of R&D projects (€2M)
- Example: autonomous-drone inspection of a **nuclear power plant**

Multi-Robot Systems @ CTU (M. Saska lab) *2016–12/2025*
Lab Assistant, Researcher & Developer

- Fundamental & applied research (see academic record)
- Field exp. (hds. of autonomous flight hours; example video)
- Technical solutions, R&D PoCs, projects leadership & admin, mentoring, industrial cooperations, public demos, teaching
- Open-source contributor to MRS UAV System

Selected projects

DARPA Subterranean Challenge (**link**)

- Subterranean search & rescue with autonomous robot teams
- Team lead of UAVs (team *CTU-CRAS-NORLAB*)
- Responsibilities: system design, GNSS-denied autonomy, mapping, SLAM, field exp., integration, and deployment in comp.
- Winning $1M while competing with top world institutions

Dronument (**video**)

- Documenting interiors of historical monuments with multiple cooperating UAVs in a synchronized multi-robot formation
- Achieved reliable operation and deployed the autonomous system in 20+ historical sites (incl. 2 UNESCO sites; see paper)

Multi-UAV swarming (**video**)

- Deployed first fully-decentralized swarms w/o communication

DOFEC (**video**)

- Fire detection and extinguishment using autonomous UAV

Education & Research

PhD, Autonomous Flying Robotics CTU, 2019–2024

- Topic: UAV autonomy in perception-degraded settings (**pdf**)
- 19 publications; h-index: 12 (WoS), 17 (GScholar)

BSc & MSc, Cybernetics & Robotics CTU, 2014–2019

Honors & Awards

1x Werner von Siemens prize (**web**, **video**) 2025
#1 dissertation out of all 2023-2024 STEM works in Czechia

1x Joseph Fourier prize (**web** in Czech) 2025
#1 dissertation out of all 2024 CS works in Czechia

1x Best paper award at IEEE RA-M (**web**, **paper**) 2025
Robotics & Automation Magazine award for our **Dronument** work

2x AI Awards (**web** + **our tech** in the EU annual report) 2025
Team award for contributions in Czech AI R&D

3x Dean's prize (**link**) 2024, 2019, 2017
All my theses (dissertation, master's & bachelor's) were selected in top 1% works at CTU in their respective category and year

1x Czech National Excellence Award M17+ 2022
Excellent international evaluation of our Dronument solution (**link**)